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[22BSMA31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

FUNCTIONAL TRANSFORMS AND NUMERICAL METHODS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Missing data may be suitably assumed.
ii. Answer All Questions

PART-A

05 X 02=10

1.	a.	If $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in $0 < x < 2\pi$, then find the Fourier coefficient a_0 of $f(x)$.	L2	CO1	02
	b.	Find the Fourier sine transform of $2e^{-3x} + 3e^{-2x}$.	L2	CO1	02
	c.	Define Correlation and Regression.	L1	CO1	02
	d.	Construct a finite difference table for the function $f(x) = x^3 + x + 1$ where x takes the values 0, 1, 2, 3, 4.	L2	CO1	02
	e.	Use Simpson's (3/8)th rule to obtain the approximate value of the integral $\int_0^{0.3} f(x)dx$ using the following table:	L1	CO1	02

x	0	0.1	0.2	0.3
$f(x)$	1	0.996	0.9675	0.8854

PART-B

06 X 05=30

2.	a.	Find the Fourier expansion for $f(x) = \|x\|$ in $-\pi \le x \le \pi$.	L3	CO2	05
		OR			
	b.	Expand $f(x) = 2x - 1$ as cosine series in $0 < x < 1$.	L3	CO2	05
3.	a.	Find the Fourier transform of $f(x) = e^{-\|x\|}$.	L3	CO3	05
		OR			
	b.	Obtain the Z-transform of $\cosh n\theta$	L3	CO3	05
4.	a.	If $\bar{u}(z) = \frac{2z^2+3z+4}{(z-3)^3}, \|z\| > 3$, find the values of u_1, u_2.	L3	CO3	05
		OR			
	b.	Find the inverse z-transform of $\frac{z}{(z-1)(z-2)}$.	L3	CO3	05
5.	a.	If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$. Show that $\sigma_y = 2\sigma_x$ or $\sigma_x = 2\sigma_y$.	L3	CO5	05
		OR			

10. a. Find the correlation coefficient and the equation of the lines of regression for the following values of x and y. L3 CO5 **10**

x	1	2	3	4	5
y	2	5	3	8	7

OR

b. Fit a parabola by the method of least squares for the following data. L3 CO5 **10**

x	2	4	6	8	10
y	3.07	12.85	31.47	57.38	91.29

11. a. From the following data estimate the number of students who have scored L3 CO6 **10**

i) less than 45 marks,

ii) Between 40 and 45 marks.

Marks	30-40	40-50	50-60	60-70	70-80
No. of students	31	42	51	35	31

OR

b. Using Newton's divided difference formula find $f(8), f(15)$ from the following data. L3 CO6 **10**

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

12. a. Evaluate $\int_0^{\frac{\pi}{2}} \cos x \, dx$ by applying Simpson's 1/3rd rule taking eleven ordinates. Compare the value with the theoretical value. L3 CO6 **10**

OR

b. Evaluate $\int_0^1 \frac{x}{1+x^2} dx$ by Weddle's rule taking seven ordinates and hence find $\log_e 2$. L3 CO6 **10**

13. a. Using Runge-Kutta method of fourth order, find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}, y(0) = 1$ taking $h = 0.2$. L3 CO6 **10**

OR

b. Apply Milne's method to compute $y(1.4)$ correct to four decimal places given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the following data: $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$. L3 CO6 **10**

b. Show that if θ is the angle between the lines of regression, then $tan\theta = \frac{\sigma_x \sigma_y}{\sigma_x^2+\sigma_y^2}\left(\frac{1-r^2}{r}\right)$. L3 CO5 **05**

6. a. Find a real root of the equation $xe^x - 2 = 0$ correct to three decimal places using Newton-Raphson method. L3 CO6 **05**

OR

b. The area of a circle (A) corresponding to diameter (D) is given below: L3 CO6 **05**

D	80	85	90	95	100
A	5026	5674	6362	7088	7854

Find the area corresponding to diameter 105 using an appropriate interpolation formula.

7. a. Use Simpson's $\frac{3^{th}}{8}$ rule to evaluate $\int_1^4 e^{\frac{1}{x}}\,dx$. L3 CO6 **05**

OR

b. Evaluate $\int_4^{5.2} \log x\,dx$ taking 6 equal strips using Weddle's rule. L3 CO6 **05**

PART-C **06 X 10=60**

8. a. If $f(x) = x(2\pi - x)$ in $0 \le x \le 2\pi$. Show that $f(x) = \frac{2\pi^2}{3} - 4\left(\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \cdots\right)$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$ L3 CO2 **10**

OR

b. Obtain the Fourier series of y up to the second harmonics for the following values: L3 CO2 **10**

x^0	45	90	135	180	225	270	315	360
y	4	3.8	2.4	2	-1.5	0	2.8	3.4

9. a. Find the Fourier transform of $f(x) = \begin{cases} 1-|x|; & |x| \le 1 \\ 0; & |x| > 1 \end{cases}$ and hence, deduce that $\int_0^\infty \frac{\sin^2 t}{t^2}\,dt = \frac{\pi}{2}$. L3 CO3 **10**

OR

b. Solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using Z-transforms. L3 CO4 **10**